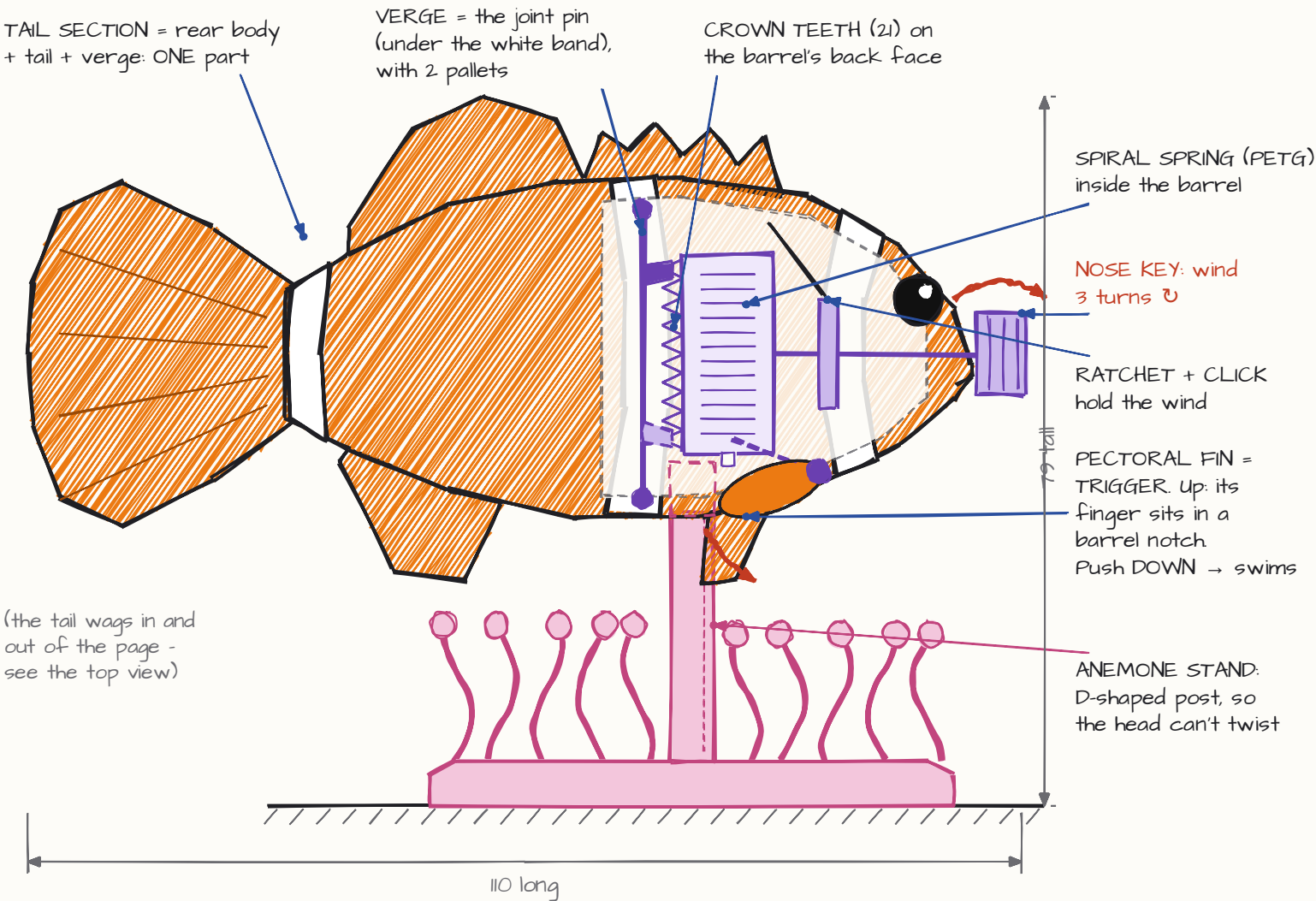


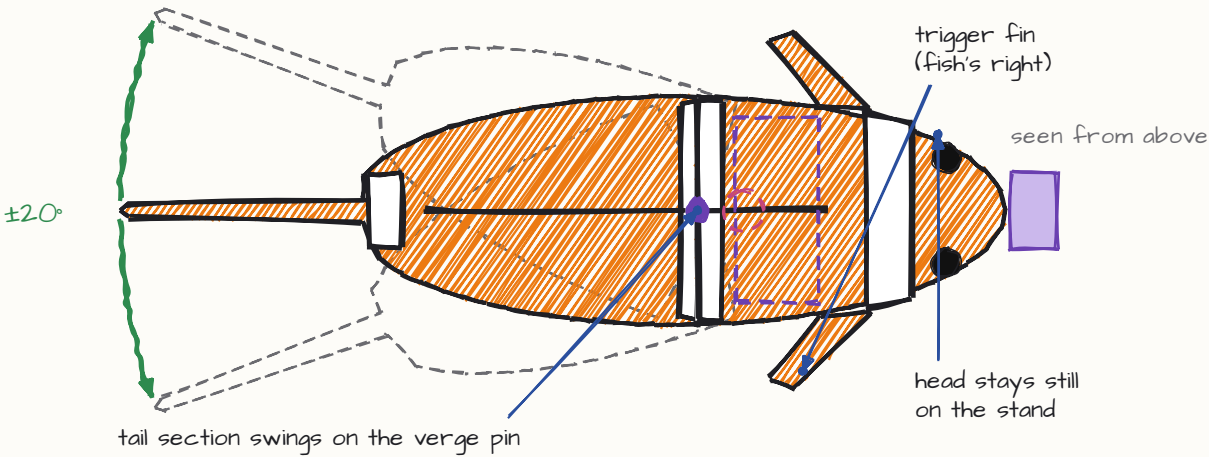
CLOWNFISH HOVER - HOW IT WORKS

spring → crown wheel → verge → the tail wags in place · purple = mechanism

SIDE VIEW - cut away · from the fish's right · scale 1.4 : 1

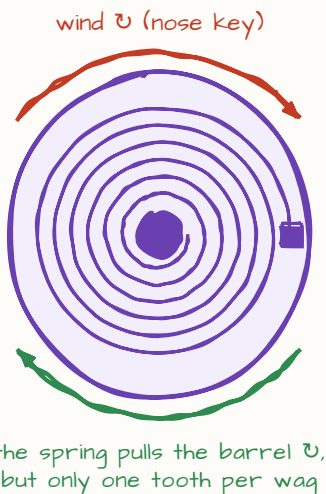


TOP VIEW - THE TAIL WAGS ±20°

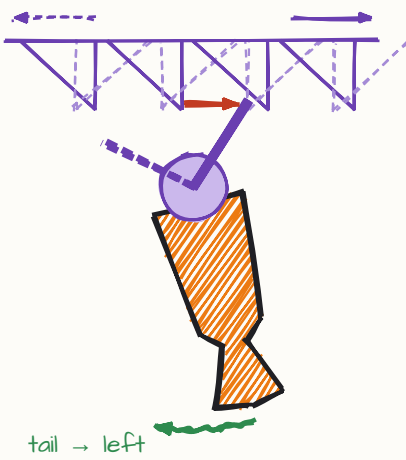


THE TRICK - A VERGE ESCAPEMENT (as in the first clocks)

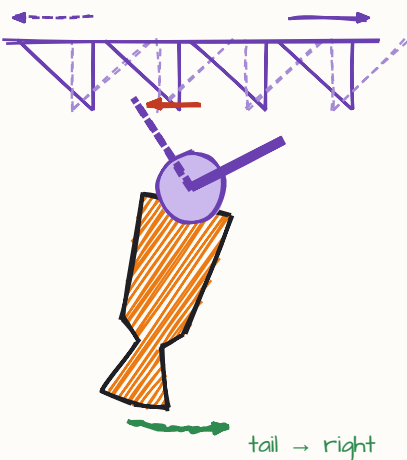
a SPIRAL SPRING + BARREL



b TOOTH PUSHES THE UPPER PALLET



c ...THEN THE LOWER ONE



- ① Winding the nose key coils the spring the click stops it running back.
- ② The spring turns the barrel, but its 21 crown teeth have to get past the two pallets.
- ③ A top tooth shoves the upper pallet → the tail swings; that swings the lower pallet into a bottom tooth → it shoves back → the tail swings back (b ↔ c).
- ④ One tooth per wag; the spring can only let go a little at a time, so it keeps going.

HOW IT PLAYS

- ① Push the fish onto its anemone stand (D-shaped post, click).
- ② Fin UP (locked). Turn the nose key 3 turns - click-click...
- ③ Push the pectoral fin DOWN → the tail wags ≈ 4× a second for ≈ 15 s, nose still.
- ④ Fin back UP stops it any time.

WHY IT LASTS ≥ 5 s

- Crown wheel 21 teeth = 21 wags per barrel turn
- Spring wound 3 turns → ≈ 63 wags
- Wag rate ≈ 4/s (the tail's weight sets it) → swims ≈ 15 s
- Even twice as fast → 7.5 s ✓
- PETG strip 0.45 × 5 mm, 10 coils: ≈ 4 Nmm wound, 25 MPa (safe); ≈ 40 mJ stored, ≈ 0.6 mJ per wag
- Too fast? A heavier tail wags slower and swims longer

PARTS (8, ALL PRINTED) + RULES

- 1 head left 2 head right (snap together)
- 3 nose key + arbor + ratchet
- 4 spiral spring (PETG)
- 5 barrel with 21 crown teeth
- 6 tail section = body + tail + verge
- 7 pectoral-fin trigger 8 anemone stand
- ✓ 100 % printed, every part fits the egg
- ✓ Built: 110 long, 79 tall → bigger than the egg
- ✓ Trigger = fin lever, no removable pin
- ✓ Opening the head lets the spring go
- ✓ Snap fits: no glue, no tools

BIOMIMICRY

Clownfish don't cruise: they hover beside their anemone, sculling with quick tail beats. This one does the same on its anemone stand, and its white bands hide the moving joint.

ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 3 · sheet 1 of 1

Concept 3: CLOWNFISH HOVER

Theme: Biomimicry - clownfish hovering at its anemone
Toy type: Spring toy, swims in place ≥ 5 s
Scale / units: mm · side view 1.4 : 1 on 11×17

Drawn by: _____ Signature: _____ Date: _____